Lab Experiments 1-7

Data Handling & Visualization: Python Code & Output

Experiment 1

Student Average Scores

PYTHON

Code

```
import numpy as np

student_scores = np.loadtxt("Students_data.csv", delimiter=",", skiprows=1, usecols=(1, 2, 3, 4
    ))
subjects = ["Math", "Science", "English", "History"]

average_scores = np.mean(student_scores, axis=0)

highest_subject = subjects[np.argmax(average_scores)]

print("Average scores:", np.round(average_scores, 2))
print("Subject with highest average score:", highest_subject)
```

Output

```
Average scores: [76.33 80.5  76.67 80.67]
Subject with highest average score: History
```

Experiment 2

Past Month Price Average

PYTHON

Code

```
import pandas as pd

df = pd.read_csv("Sales_data.csv")
average_price = df[df["Month_sales"] == "November 2023"]["Price"].mean()

print("Average price in November 2023 (Past Month): ", round(average_price, 2))
```

Output

```
Average price in November 2023 (Past Month):  221.78
```

Experiment 3

House Average Price

PYTHON

Code

```
import numpy as np

house_data = np.loadtxt("House_data.csv", delimiter=",", skiprows=1)

houses_more_than_4 = house_data[house_data[:, 1] > 4]
average_price = np.mean(houses_more_than_4[:, 5])

print("Average price of houses with more than 4 bedrooms: ", round(average_price, 2))
```

Output

```
Average price of houses with more than 4 bedrooms:  409598.18
```

Experiment 4

Quarterly Sales Analysis

PYTHON

Code

```
import numpy as np

sales_data = np.genfromtxt("Sales_data.csv", delimiter=",", skip_header=1, dtype=str)

months = sales_data[:, 1]
sales = sales_data[:, 4].astype(float)

Q1 = 0.0
Q4 = 0.0

for i in range(len(months)):
    if ("January" in months[i]) or ("February" in months[i]) or ("March" in months[i]):
        Q1 += sales[i]
    elif ("October" in months[i]) or ("November" in months[i]) or ("December" in months[i]):
        Q4 += sales[i]

total_sales_year = np.sum(sales)

percentage_increase = ((Q4 - Q1) / Q1) * 100

print("Total sales for the year:", round(total_sales_year, 2))
print("Percentage increase from Q1 to Q4:", round(percentage_increase, 2), "%")
```

Output

```
Total sales for the year: 16884.79
Percentage increase from Q1 to Q4: 54.37 %
```

Experiment 5

Fuel Efficiency Analysis

PYTHON

Code

```
import numpy as np

fuel_data = np.genfromtxt('Fuel_data.csv', delimiter=',', skip_header=1, dtype=str)

fuel_efficiency = fuel_data[:, 7].astype(float)
make = fuel_data[:, 8]

average_efficiency = np.mean(fuel_efficiency)
mazda_eff = fuel_efficiency[make == "mazda"]
audi_eff = fuel_efficiency[make == "audi"]
mazda_avg = np.mean(mazda_eff)
audi_avg = np.mean(audi_eff)

percentage_improvement = ((mazda_avg - audi_avg) / audi_avg) * 100

print("Average Fuel Efficiency of all cars:", round(average_efficiency, 2), "MPG")
print("Average Fuel Efficiency of Mazda:", round(mazda_avg, 2), "MPG")
print("Average Fuel Efficiency of Audi:", round(audi_avg, 2), "MPG")
print("Percentage Improvement from Mazda to Audi:", round(percentage_improvement, 2), "%")
```

Output

```
Average Fuel Efficiency of all cars: 29.33 MPG
Average Fuel Efficiency of Mazda: 30.1 MPG
Average Fuel Efficiency of Audi: 29.03 MPG
Percentage Improvement from Mazda to Audi: 3.67 %
```

Experiment 6

Customer Purchase Analysis

PYTHON

Code

```
import pandas as pd

df = pd.read_csv("grocerystore.csv")
total_sales = df["Sales"].sum()
print("Total sales for the grocery store: ", round(total_sales, 2))

discount = 0.10
tax = 0.18
price_after_discount = total_sales - (total_sales * discount)
price_after_tax = price_after_discount + (price_after_discount * tax)

print("Final amount after applying discount and tax: ", round(price_after_tax, 2))
```

Output

```
Total sales for the grocery store:  13948.83
Final amount after applying discount and tax:  14813.66
```

Experiment 7

Customer Order Analysis

PYTHON

Code

```
import pandas as pd

df = pd.read_csv("order_data.csv")
total_orders = df.groupby("Full Name")["Order Count"].sum()
avg_order_per_product = df.groupby("Items")["Order Total"].mean()
earliest_order = df["Order"].min()
latest_order = df["Order"].max()

print("Total number of orders made by each customer: ", total_orders)
print("Average order quantity for each product: ", avg_order_per_product)
print("Earliest order date: ", earliest_order)
print("Latest order date: ", latest_order)
```

Output

```
Total number of orders made by each customer:  Full Name
Aarav Sharma    22
Anika Rao       15
Diya Patel      28
Kabir Khan      19
Meera Iyer      23
Rohan Gupta     25
Sara Reddy      28
Vikram Nair     21
Name: Order Count, dtype: int64
Average order quantity for each product:  Items
Headphones    600.880909
Keyboard      793.656250
Laptop        721.737143
Monitor       763.680909
Mouse         812.640000
Tablet        971.296250
Webcam        924.258889
Name: Order Total, dtype: float64
Earliest order date:  2023-01-05
Latest order date:  2023-12-31
```